N83 Q0: reconciling A (rho faster) and B (IC faster)

Status (2026-10-15): measured. GitHub run 38617911080 (commit `4ab2a678`, one `ubuntu-24.04` runner) completed the frozen protocol: all 284 rows ran and were valid. The receipt is archived in `run_github_38617911080/`, and `verify_receipt.py` re-run on it reproduced `result.json` and `result_table.md` byte for byte. See Result. The design sections below are the protocol as frozen before the run.

Two measurements of the same point disagree in direction:

	A	B
Where	crypto `research/n83_evaluation_20261012/`	cryptanalysis `experiments/koblitz-single-target-n83-tie` (run 38523707244)
Crypto source	`a8194d61`	`770f6a69`
Rho	`StrongRhoG<WideGf2,u64>` default (32 lanes, DP 4, escape off), seeds 202610061..65	`koblitz_rho_batch_ks_strong_online` rung 3 (32 lanes, DP 4), seed 831000
IC	`koblitz_orbit_dlp_fast`, fresh in-process rank (seeds 21..25)	`koblitz_orbit_dlp_fast_online` + `ic_log_cache.patch`, cached factor-log table
Host	5-vCPU KVM guest, one pinned CPU	GitHub ubuntu-24.04 runner, one pinned CPU
Q0 result	IC/rho online 1.382 [1.036, 1.882] (5 pairs)	rho/IC online 1.779 [1.775, 1.783] (8 pairs, one rho walk)
Reading	rho online faster	IC online faster

Point Q0 = (265589834553716618053296 0, 388152630799167326529261 5) on `EC1N83Ce1h35c7ced6b497` (y^2 + xy = x^3 + x^2 + 1 over GF(2)[z]/($z^{83+z}7+z^{4+z}2+1$), r = 8569786107849059), retained K600 base `d4a4da26...a9f7`.

Result

Frozen reading: `decided`. A cell `A_not_reproduced`, B cell `B_reproduces`. Responsible: `host_or_instance` ("A's direction does not survive the move to this runner"). All 284 rows were valid. The A/A controls passed (rho +0.000 [-0.004, 0.004]; IC +0.009 [-0.006, 0.024]), as did the positive control (+0.007 [-0.002, 0.014] from the exact 0.978) and every identity check. One row (iA770f, block 1) carried a contention flag (major page faults). The decision recomputed without flagged blocks is the same.

Quantity on Q0 (ln rho/IC; 4 blocks)	mean	ratio	95%	class
L_A, A's exact configuration	+0.091	1.095	[0.071, 0.107]	
rev_rho (a8194d61 to 770f6a69)	-0.009	0.991	[-0.012, -0.004]	negligible
impl_rho (rung 3 vs StrongRhoG, Shapley)	+0.299	1.348	[0.294, 0.303]	contributes
seed (831000 vs A's five, Shapley)	+0.184	1.202	[0.179, 0.190]	contributes
boundary	-0.000	1.000	[0.000, 0.000]	negligible
rev_ic	-0.017	0.983	[-0.043, -0.002]	negligible
prod_ic	+0.016	1.016	[-0.014, 0.055]	unresolved
cache	+0.001	1.001	[-0.007, 0.008]	negligible
L_B, B's exact configuration	+0.564	1.757	[0.550, 0.574]	

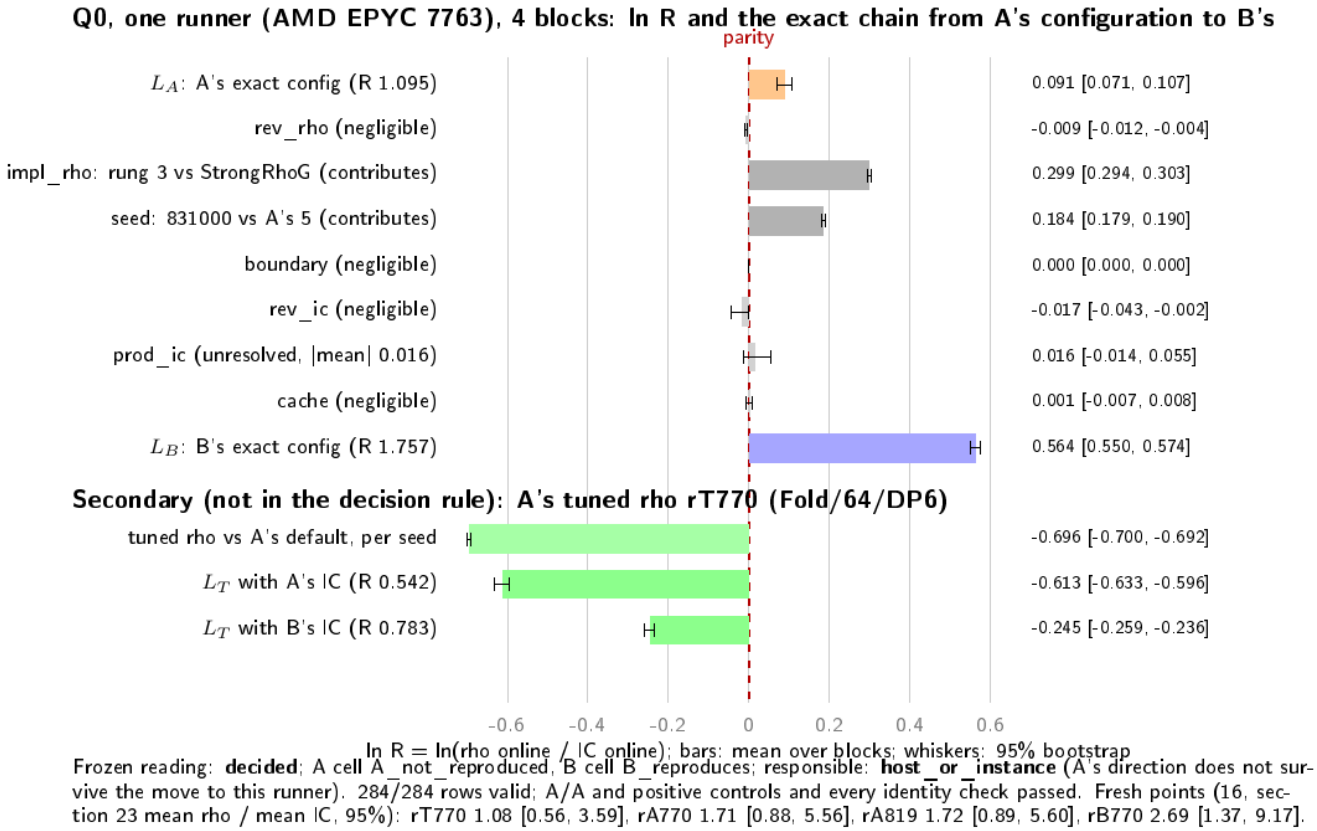


Figure 1: Result of run 38617911080

Figure 3. The frozen chain and the secondary tuned-rho readings (source, vector); numbers from `run_github_38617911080/panel/r`

What it means, plainly.

- On this runner, A's own configuration already reads IC 1.10x faster online. So A's "rho faster" (1.38 in IC's disfavour) did not survive the move to this runner. The responsible factor under the frozen rule is therefore the host/instance, not any code factor. Here A's IC producer ran at 505 ns per probe (715 on A's host), while A's StrongRhoG ran at 490 ns per step (461 on A's host). The residual between A as published and A on the runner is -0.415 in ln R.
- This runner was an **AMD EPYC 7763**, not the EPYC 9V74 that A's host and B's runner reported. "Host or instance" therefore includes a CPU model change. The run cannot say which property of A's host made its IC relatively slow.
- B's configuration reproduces (1.757, against B's published 1.779). Most of B's larger lead on Q0 comes from the rho it was measured against:
  - **A weaker rho.** Rung 3 costs 618 ns per step against StrongRhoG's 490 on the same CPU (+0.229 in ln time at A's seeds).
  - **An unlucky seed.** Under seed 831000, rung 3's walk took 12,804,216 steps against a geometric mean of 8.41M over A's five seeds (+0.42 within rung 3). StrongRhoG at the same seed took only 9,413,926.
  - Together (Shapley) this is +0.299 for the implementation and +0.184 for the seed. IC-side code differences (revision, producer, log cache) are each under 0.02 in mean.
- **Against A's tuned rho (rT770, Fold/64/DP6), IC shows no online advantage.** This is a secondary reading, not part of the frozen decision. On Q0 the tuned rho is 0.50x the default per seed. Rho/IC is 0.542 [0.531, 0.551] with A's IC (L_T -0.61) and 0.783 [0.772, 0.790] with B's cached IC (L_T -0.24): rho faster in both. On B's 16 fresh points, section 23 mean rho / mean IC gives:

Rho arm	mean rho / mean IC [95%]
rT770 (tuned)	1.08 [0.56, 3.59] (not separated from parity)
rA770	1.71 [0.88, 5.56]
rA819	1.72 [0.89, 5.60]
rB770 (B's rung 3)	2.69 [1.37, 9.17]

- The B panel's online IC advantage (1.78x on Q0, 2.77x on 16 points) is real against the rung-3 reference it used. It came from a rho that is slower per step than the library's StrongRhoG, and on Q0 also from a seed that gave rung 3 a long walk. Against the tuned rho it disappears on Q0 and is not separated from parity over the fresh points.

Audit predictions (SOURCE_AUDIT section 7) against the outcome:

- Predictions 1 (rev_rho negligible, identical steps), 4 (boundary about 1e-4) and 6 (rung 3 slower per step, tuned rho fastest on fresh points) held.
- Prediction 2's size held (impl_rho 0.299), but "sufficient to flip" was not evaluable, because A did not reproduce.
- Prediction 3 (seed about +0.25) came out at +0.184 (Shapley).
- Prediction 5's IC terms each have means under 0.02, but prod_ic's interval reaches 0.055, so it is unresolved. Its L_A forecast (about -0.10) was wrong: L_A was +0.091.
- The fresh-point seed-family negative control is -0.24 [-0.47, 0.007]. Its interval only just includes 0. Seed relabelling moves single walks by large factors, and 16 points are few.

Limits.

- One runner, one instance; Tier 1 instance scope.
- Q0 intervals cover repeat and timing variation only.
- The IC factors were measured on Q0 only.
- The host effect is not split further.
- The schedule finished 3.3 h after job start (estimate 4.1-5.3 h); no segment was skipped.

Source audit (before any timing)

SOURCE_AUDIT.md (PDF). Both IC runs did identical work on Q0: 8,845,441 probes and the same relation, at every rank seed, cached or fresh. So the gap splits exactly into rho steps x ns per step over IC probes x ns per probe:

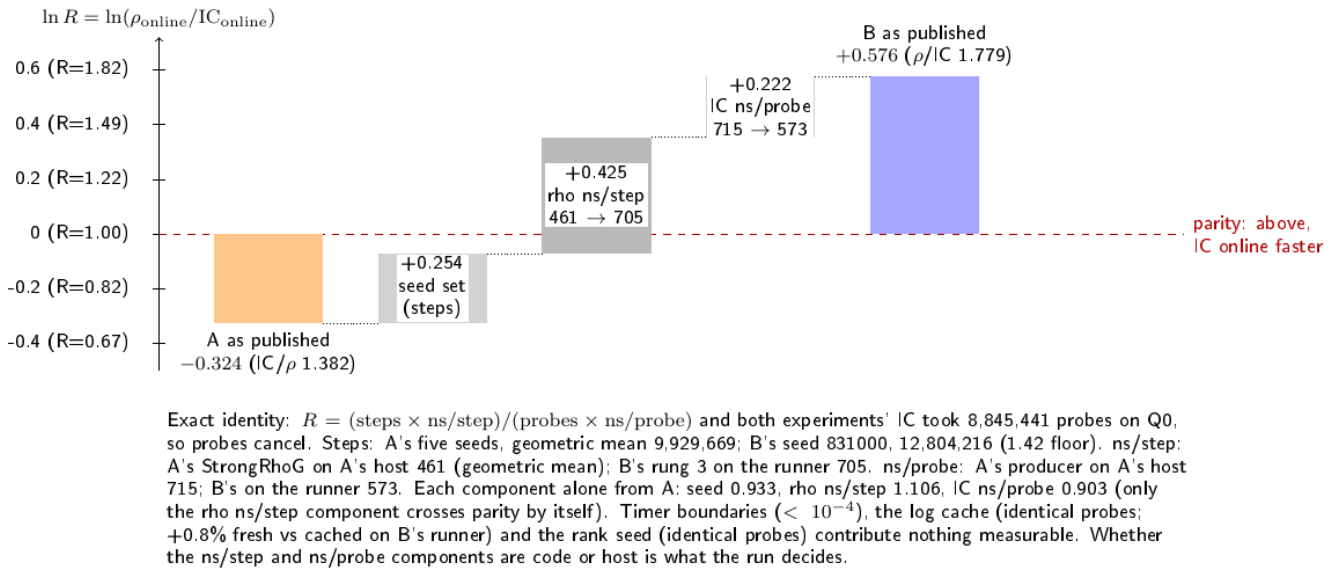


Figure 2: Exact decomposition of the published gap

Figure 1. The published A to B gap in ln(rho/IC) on Q0, split into its three exact components from the raw receipts (source, vector). These are not one-host measurements.

- **Explained without a run:** the two harnesses' timer boundaries cover the same operations (difference under 1e-4 in ln R). The log cache moves no work (identical probes; +0.8% fresh against cached on B's runner). The rank seed is irrelevant online. A's median agrees with A's geometric mean. B's seed made Q0 a long walk: 1.42 times the floor, against a geometric mean of 1.10 over A's five seeds. That gives +0.254 in ln R, which cannot flip A's direction alone (0.933).
- **Not explained without a run:** rho ns per step is 461 for A's StrongRhoG on A's host and 705 for B's rung 3 on the runner (+0.425, enough to flip alone: 1.106). IC ns per probe is 715 for A's producer on A's host and 573 for B's on the runner (+0.222). Each could be code or host: both hosts report an AMD EPYC 9V74 and rustc 1.99.0, but they are different VMs.

Frozen design (protocol.json, schedule.json)

Figure 2. Arms and the exact chain from A's configuration to B's (source, vector).

Arm	Code	Revision	Role
rA819	A's `n83_strong_point_eval` (StrongRhoG default)	a8194d61	A's exact rho
rA770	same adapter	770f6a69	rho revision
rA770aa	same as rA770	770f6a69	A/A control
rB770	rung 3	770f6a69	B's exact rho
rT770	A's tuned worker, Fold/64/DP6	770f6a69	reported, not in the chain
iA819f	A's `koblitz_orbit_dlp_fast`, fresh sequential rank	a8194d61	A's exact IC
iA770f	same producer	770f6a69	IC revision

Arm	Code	Revision	Role
iB770f	B's online producer + patch, fresh sequential rank	770f6a69	IC producer
iB770c	same, cached table (`KIC_LOG_TABLE_IN`)	770f6a69	B's exact IC
iB770caa	same as iB770c	770f6a69	A/A control

- **Points:** Q0, plus B's 16 fresh public points (B's `inputs/targets.json`, hash-bound).
- **Q0 panel:** 4 complete blocks. Each block runs every rho arm at all six seeds (A's 202610061..65 and B's 831000) and every IC arm once: 35 runs, in a frozen pseudo-random order. Fresh-rank arms use rank seed 20 + block.
- **Fresh panel:** per point, rA819, rA770, rB770 and rT770 at seeds 830000 + w (B's rule) and 202610061 + w (A's family), and one cached IC run. That is 9 runs per point.
- **Order:** Q1, F(w0-3), Q2, F(w4-7), Q3, F(w8-11), Q4, F(w12-15); 284 runs in all. One process at a time on one pinned CPU, in one job.
- **Factorial:** every arm runs in every block, so every rho arm x seed x IC arm combination is a within-block paired ratio. Rho implementation (3 levels) x seed set (2) is fully crossed. Revision is nested in A's code, and the log source in B's IC producer, because the other code does not exist at n=83 at a8194d61 (SOURCE_AUDIT section 5).
- **Statistic:** natural logs. Per block, `L_A` = mean over A's seeds of ln rA819 minus ln iA819f; `L_B` = ln rB770[831000] (B's clock) minus ln iB770c. Then `L_B - L_A` = rev_rho + impl_rho + seed + boundary + rev_ic
  - prod_ic + cache, exactly. impl_rho and seed are Shapley values over the 2x2. The bootstrap resamples blocks (10,000 draws; 95% intervals, and Bonferroni 99.29% for m = 7). On the fresh points: each rho factor per point, bootstrapped over points, and B's section 23 statistic per rho arm.
- **Decision rule:** a factor is *sufficient to flip* if A reproduces on the runner (L_A upper bound < 0) and L_A + D_f has a Bonferroni lower bound > 0. It *contributes* if its Bonferroni interval excludes 0 and $|D_f| >= 0.05$. It is *negligible* if its 95% interval lies inside [-0.05, 0.05]. The responsible factors are every sufficient one, or else the smallest greedy set of contributing ones that flips L_A. If A's configuration already reads IC-faster on the runner, the responsible factor is `host_or_instance`.
- **Controls:** two A/A twins (rho on all six seeds, and cached IC). Each passes if $|\text{mean ln}| <= 0.03$ and its 95% interval lies inside [-0.05, 0.05]. A positive control: the instrument must recover, within 0.10, the exact ln step ratio 0.978 between rA770's seeds 202610064 and
202610063. Identity checks: one scalar per point, equal to Q0's sidecar and replayed in `oracle.py`; identical steps per seed across rA819, rA770 and rA770aa; identical probes across IC arms; every fresh table's logs equal to B's archived table.
- **Validity:** invalid rows (exit, verification, timeout, memory event) make `no_result`. Contention flags from isolated_bench's Tier 1 gate (steal, throttling, major faults) keep the row. The decision is recomputed without flagged blocks, and a different answer reads `sensitive_to_contention`.

Resources and expected runtime

Stage	Estimate	Basis
rustup 1.99.0, two crypto builds, bin tests	15-30 min	local builds 5.6 + 4.7 min at -j3
B's table re-verified in oracle.py	about 10 s	local
Q0 block (3 fresh-rank IC at about 14-18 min each, 2 cached IC, 30 rho)	48-62 min	B runner: fresh B producer 888 s, cached 52 s, rung 3 705 ns per step; A host: A's IC 1,054 s per process
Fresh segment (4 points x 9 runs)	6-8 min	B Panel B: IC median 645 ms online + 46 s setup, rho 6.2 s
Audit	5-15 min	oracle replays and log-table checks
Total	**about 4.1-5.3 h**	job timeout 350 min; a time guard skips whole segments past 325 min

Memory: fresh-rank IC peaks at about 5.8 GB, one process at a time, within 16 GB.

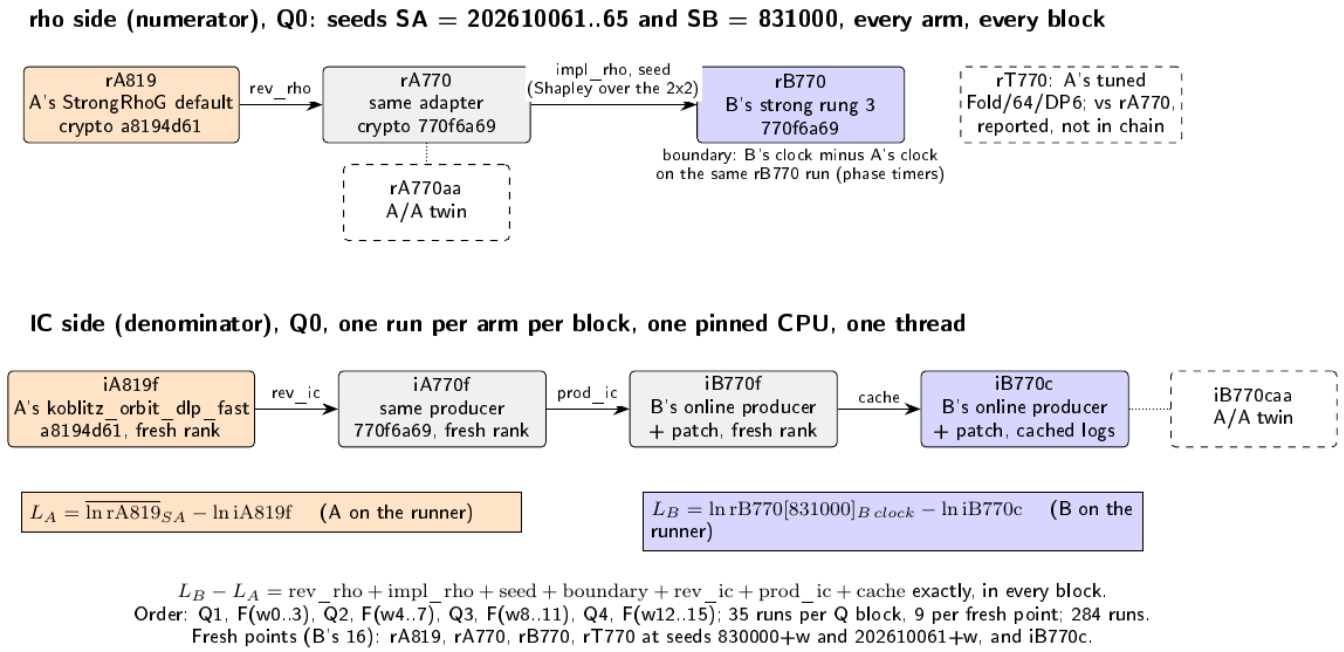


Figure 3: Arms and the seven-term chain

Local validation (correctness only; no timing is reported)

Local validation ran in a shared, contended container. Receipts: `smoke/`.

- `restore_sources.py`: both snapshots match `protocol.json`. Every file shared with B also matches B's frozen digests, and A's adapter matches A's recorded 82bc92d2.
- Both crates build with `--locked`. `cargo test --release --locked --bins` passes: 12 tests (r819) and 32 tests (r770).
- `run_schedule.py --segments Q1,F1 --one-per-arm` ran all 10 arms on Q0 and the 5 fresh-panel arms on point w00: 15 valid rows. Then `verify_receipt.py --correctness-only` passed every identity check (see the list after this one).
- Earlier direct `arm.py` calls reproduced A's and B's exact step counts on Q0: 10,203,425 at seed 202610061 for rA819 and rA770, and 12,804,216 at seed 831000 for rB770.
- `make_schedule.py --check` passes. `verify_log_table.py` passes on B's archived table. `verify_receipt.py` on synthetic data reproduced the chain identity to rounding, and a deadline-0 run of `run_schedule.py` recorded all 284 runs as `skipped_time_guard`.
- Workflow: YAML parses; `actionlint` 1.7.7 with shellcheck 0.10.0 passes.

The identity checks that passed:

- one scalar per point; Q0's equals the sidecar 3141592653589793; every scalar replays in `oracle.py`;
- steps equal A's receipts (14,229,982 and 13,159,092 at seeds 202610064 and 202610065) and B's (9,447,525 on w00 at seed 830000);
- probes are 8,845,441 for every IC arm on Q0 and 14,455,639 on w00, as in B's Panel B;
- the fresh B-producer table's 600 logs equal B's archived table.

Gaps and limits

- The chain is measured on **one runner**. A residual between A's published L_A and the runner's L_A is the A-host effect, and it cannot be split further here.
- B's IC producer and rung 3 have no n=83 path at a8194d61. Their revision effects are therefore not estimable, and only A's code carries the revision factor.
- IC producer, revision and log-source effects are measured on Q0 only (each fresh-rank arm costs about 15 min). The fresh points carry the rho factors over targets.

- Seed-set effects are fixed effects of these seeds. Q0 intervals cover repeat and timing variation, not seed or target variation.
- Tier 1 instance scope: one pinned exposed CPU on a shared VM. Exclusivity is not certified.
- Canonical graphs checked: B's `figures/prior_ratios.*` and `figures/tier1_result.*`, and `experiments/volcano-ic/fig`. They are left unchanged because they plot B's frozen numbers, which reproduce here. This directory's figures are new, and the correction is carried in `papers/frobenius-orbits`.

Files

`protocol.json` (frozen design and digests), `schedule.json` (frozen order), `SOURCE_AUDIT.md`, `restore_sources.py`, `crates/{r770,r819}/`, `arm.py`, `make_schedule.py`, `run_schedule.py`, `verify_log_table.py`, `verify_receipt.py`, `figures/`, `smoke/`, `run_github_38617911080/` (the run's artifact), and the workflow `.github/workflows/n83-reconcile.yml`. The workflow reuses from B: `ic_log_cache.patch`, `inputs/targets.json` and the archived factor-log table of run 38523707244.

Reproduce

```
git clone https://github.com/aburan28/crypto /path/crypto
python3 restore_sources.py --crypto-repo /path/crypto
python3 make_schedule.py --check
(cd crates/r819 && cargo build --release --locked); (cd crates/r770 && cargo build --release --locked)
python3 verify_log_table.py --table ../koblitz-single-target-n83-tier1-20261014/tier1_github_38523707244/pr
python3 run_schedule.py --bin-dir $CARGO_TARGET_DIR/release \
  --log-table ../koblitz-single-target-n83-tier1-20261014/tier1_github_38523707244/precompute/factor_logs_n8
  --receipt-dir R/panel --cpu 0
python3 verify_receipt.py R/panel
```